

Progress Test 1.B — Elec-S1

1.B · From charge to electric current

Semester 1 · Academic Year 2026–2027

ANSWER KEY · MODEL ANSWERS AND MARKING SCHEME

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion

(1 mark)

The capacity of a phone battery is $Q = 4000 \text{ mA h}$, i.e. the charge it delivers when a current of 4000 mA flows for one hour. Express Q in coulombs. Give the exact result.

$Q = 4000 \times 10^{-3} \text{ A} \times 3600 \text{ s} = 4.000 \text{ A} \times 3600 \text{ s}$	(and $1 \text{ A s} = 1 \text{ C}$)
$Q = 14\,400 \text{ C} = 1.44 \times 10^4 \text{ C}$	

Marking

①
0 1
☐ ☐

Marking scheme

① 1 pt $1.44 \times 10^4 \text{ C}$ (the hour converted into 3600 s).

Examiner's tip

The coulomb is a *derived* unit, $1 \text{ C} = 1 \text{ A s}$: a charge is a current multiplied by a time. That is why a “mA·h” is a unit of charge.

Exercise 2. Microscopic expression of the current

(5 marks)

A conductor of cross-sectional area S contains n mobile charge carriers per unit volume, each of charge q , all moving with the drift velocity v along the axis of the conductor. Express the charge dq that crosses S during dt (explain your reasoning), deduce the current i , and check that your result is dimensionally consistent, stating the SI unit of n .

During dt each carrier moves by $dx = v dt$. The carriers that cross S are those contained in the cylinder of base S and length $v dt$, of volume $S v dt$:

$$dN = n S v dt \quad \Rightarrow \quad dq = q dN = n q S v dt$$

$$i = \frac{dq}{dt} = n q v S$$

 (for electrons, $q = -e$: $i = -nevS$)

Units: n in m^{-3} ; $[n q v S] = \text{m}^{-3} \times \text{C} \times \text{m s}^{-1} \times \text{m}^2 = \text{C s}^{-1} = \text{A}$ (ampere). ✓

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Marking scheme

- ① **2 pts** Cylinder of volume $S v dt$ (1 pt) $\Rightarrow dq = n q S v dt$ (1 pt).
 ② **2 pts** $i = dq/dt$ (1 pt) $= n q v S$ (1 pt).
 ③ **1 pt** n in m^{-3} and $n q v S$ in $\text{C s}^{-1} = \text{A}$.

Examiner's tip

Test 1, Ex. 2: "current = C s^{-1} " did not earn the unit mark. The expected answer was the **ampere (A)**, one of the seven SI base units. Writing $\text{C s}^{-1} = \text{A}$ is a good dimensional check, but it does not name the unit.

Exercise 3. Current and ammeters

(3 marks)

In a copper wire carrying a current, the conduction electrons move:

- ☐ in the conventional direction of the current;
☒ in the direction opposite to the conventional direction of the current;
☐ at a speed close to that of the electric signal;
☐ only when the wire is heated.

Why: the conventional direction is that of positive charges. Electrons carry $-e$, so they move the other way ($i = -nevS$). Their drift speed is tiny compared with the speed of the signal.

An ammeter is connected:

- ☐ in parallel across the component, and has a very large internal resistance;
☒ in series in the branch, and has a very small internal resistance;
☐ in series in the branch, and has a very large internal resistance;
☐ in parallel across the component, and has a very small internal resistance.

Why: the current to be measured must flow *through* it (the circuit is opened and the ammeter inserted), and it must not change the resistance of the branch: $R_A \rightarrow 0$.

Which of these units is one of the seven SI base units?

- ☐ coulomb (C) ☐ volt (V) ☒ ampere (A) ☐ ohm (Ω)

Why: the ampere is a base unit; $\text{C} = \text{A s}$, $\text{V} = \text{W A}^{-1}$ and $\Omega = \text{V A}^{-1}$ are derived units.

Exercise 4. Counting electrons

(5 marks)

A steady current i flows through a wire during a time Δt . Determine the number N of electrons that cross a cross-section of the wire during Δt .

$$i = 3.2 \text{ mA}$$

$$\Delta t = 5.0 \text{ s}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

Steady current: $q = i \Delta t$ (from $i = dq/dt$ with i constant). Each electron carries a charge of magnitude e , so $q = Ne$:

$$N = \frac{i \Delta t}{e} \quad \text{with } i = 3.2 \text{ mA} = 3.2 \times 10^{-3} \text{ A}$$

$$N = \frac{3.2 \times 10^{-3} \text{ A} \times 5.0 \text{ s}}{1.6 \times 10^{-19} \text{ C}} = \frac{3.2 \times 5.0}{1.6} \times 10^{-3+19} = 10 \times 10^{16}$$

$$N = 1.0 \times 10^{17} \quad N \text{ is a pure number: } \text{A s C}^{-1} = 1.$$

Marking

①	0	1
②	0	1
③	0	1
④	0	1
⑤	0	1

Marking scheme

- ① 1 pt Literal expression $N = i \Delta t / e$, letters only.
- ② 1 pt Conversion $3.2 \text{ mA} = 3.2 \times 10^{-3} \text{ A}$.
- ③ 1 pt Numerical expression written *with units*.
- ④ 1 pt $N = 1.0 \times 10^{17}$.
- ⑤ 1 pt No unit for N (units cancel), two significant figures.

Examiner's tip

Without a calculator, handle the mantissas and the powers of ten separately. In Test 1 the slip was $\frac{1}{3.2 \times 10^4}$: it equals $\frac{1}{3.2} \times 10^{-4} \approx 0.31 \times 10^{-4} = 3.1 \times 10^{-5} \text{ m s}^{-1}$, not 3.2×10^{-3} .

Your score: / 14 → mark out of 20 = score $\times$ 20/14 = / 20